

When ISM 1.0 was formulated, the 28nm node was widely considered the optimal starting point for India. When do you expect India to have operational 28nm production for domestic chip designers?

The more mature nodes above 28nm should enter commercial production by 2028, while 28nm production is expected by 2029.

So, India can realistically expect indigenous commercial production at the 28nm node by around 2029?

Yes, absolutely.

New fabs typically face high initial production costs, while global competition demands competitive pricing, high yields, and quick turnaround times. When Tata begins production in 2028-29, do you believe it will be able to match international standards in quality, turnaround time, and cost?

Yes, I am confident it will, for two reasons.

First, government incentives cover up to 70% of the capital expenditure, leaving companies to fund only the remaining 30%. This significantly reduces their financial burden and places them on a level playing field with long-established global fabs.

Second, Tata, its engineering team, and its technology partner, PSMC, have the expertise needed to meet global standards. Wafer costs depend primarily on manufacturing yield and operating expenditure (OpEx). I am confident Tata will rapidly achieve yields comparable to those of PSMC. The company is also broadening its technology base through collaborations with global leaders, including Analog Devices Inc. (ADI).

India also enjoys structural cost advantages. Manpower accounts for roughly 13% of a fab's operating costs and electricity around 17%, both of which are more competitive than in many countries, including China. In addition, key raw



(L to R) Union Minister of Electronics and IT, Ashwini Vaishnaw, Gujarat CM Bhupendra Patel, PM Narendra Modi, Sanjay Mehrotra, CEO of Micron Technology, Deputy CM of Gujarat, Harsh Sanghavi at the inauguration of Micron's ATMP facility in Sanand, Gujarat, on 28th February 2026

materials such as speciality chemicals and industrial gases are already available domestically and can be refined to semiconductor-grade quality, reducing procurement costs by an estimated 5% to 7%.

Production costs may be higher during the first year or two as operations stabilise and yields improve. However, once the production line is fully optimised, there is no structural reason why Tata's wafer costs cannot be globally competitive.

Packaging has been one of the biggest successes of the ISM. Excluding Micron, what packaging capabilities can we expect from Indian projects by 2027?

By the end of 2027, Indian facilities will offer wire-bond, flip-chip, and the first generation of advanced packaging. In 2028, a dedicated 3D glass substrate facility will begin commercial production, significantly expanding our advanced packaging capabilities.

By then, Tata Electronics will also have established its packaging facility and introduced multiple

advanced packaging technologies. The company has already signed an MoU with Intel Corporation for technology transfer in advanced packaging.

Together, these developments will provide a comprehensive portfolio ranging from wire-bond and flip-chip to advanced packaging. We are also expanding the ecosystem through projects such as the Intel-3D Glass Solutions initiative, which includes a US\$3.3 billion advanced packaging facility under Semicon India 2.0. While 3D Glass Solutions is initially setting up a smaller facility in Bhubaneswar, the larger outsourced semiconductor assembly and test (OSAT) project with Intel will firmly position India on the global advanced packaging map. We are also in discussions with several other global companies, and I believe packaging will become one of India's greatest strengths.

Many chip design companies require standard ball grid array (BGA) and flip-chip BGA packaging. By 2027, how many approved